

PoseGuru: Landmarks for Explainable Pose Correction using Exemplar-Guided Algorithmic Recourse

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Abstract

Human pose correction is crucial in fields such as fitness, rehabilitation, and sports. Despite recent advancements, existing approaches often lack explainable, personalized, and fine-grained corrections. We present *PoseGuru*, an explainable optimization-based approach, leveraging algorithmic recourse and counterfactuals to iteratively refine pose landmarks by minimizing classification loss, enforcing user-specific anatomical constraints, and precisely aligning with the target pose. Our method is simple, interpretable, and adaptable, enabling easy incorporation of application-specific constraints. For robust evaluation, we introduce two new datasets, *YogaHPC* and *Pilates32+P*, generated by biomechanically perturbing landmarks of correct poses. *PoseGuru* consistently outperformed existing methods on both datasets, as assessed using metrics such as *MPIAD* and *PCIK*. Furthermore, a comprehensive user study involving yoga and pilates experts confirmed *PoseGuru*'s effectiveness, highlighting its capacity to facilitate user-driven pose correction across diverse pose types. Overall, *PoseGuru* provides an explainable and personalized solution suitable for critical applications in fitness and rehabilitation.

1. Introduction

Human Pose Correction (HPC) is the task of identifying and correcting local misaligned body parts in the pose of an individual attempting a desired target pose. Pose correction has significant real-world applications across several domains including fitness, rehabilitation, martial arts, dance forms, and sports as shown in Fig. 1, where real-time feedback on human pose can substantially aid users in refining their skills and accelerating their progress. Following are a few examples illustrating why HPC is an important problem to



Figure 1. **HPC Applications:** Kungfu, Karate, Physiotherapy, Capoeira, Yoga, Airport marshaling, Football, Ballet, Bharatanatyam. Poses are highlighted by the skeletal overlay of landmarks in red.

solve: (i) *Fitness*: Activities like yoga, pilates, resistance training, and aerobics are most effective with proper form. In weightlifting, correcting poses during squats or deadlifts ensures optimal load distribution and reduces strain, especially on the lower back. (ii) *Rehabilitation*: Rehabilitation exercises must be performed with correct form and alignment to target specific muscle groups affected by injury or surgery. Proper correction ensures effective recovery and prevents further complications. (iii) *Sports*: In sports like martial arts and gymnastics, where precise body alignment is crucial, real-time feedback-based pose correction systems significantly enhance athletic performance.

An effective pose correction system can ideally leverage Human Pose Estimation (HPE) methods to classify a human pose and identify any errors. It then suggests corrective actions to the user, thereby improving their pose and performance. HPE is a well-researched problem in computer vision and serves as a cornerstone for numerous downstream applications [8, 50]. It focuses on predicting the locations of keypoints, such as joints, on a human body in an image or video frame. Accurate keypoint localization is crucial for tasks like action recognition, which differentiates between

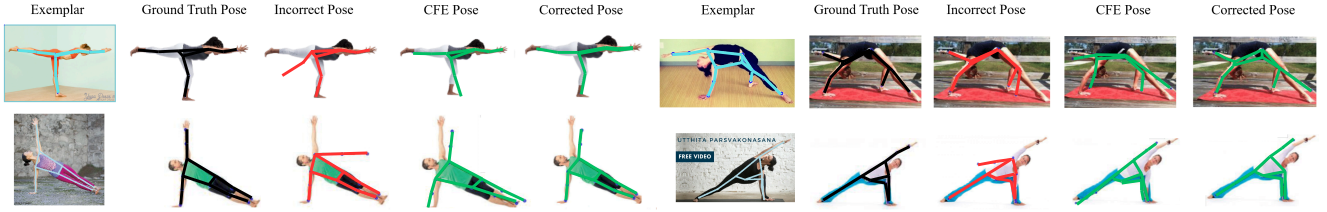


Figure 2. PoseGuru: Sample qualitative results on poses from YogaHPC and Pilates32+P datasets, including the exemplar, ground truth pose (in black), incorrect pose (in red), PoseGuru-generated counterfactual explanation (CFE) (in green) and finally the corrected pose (also in green). (While the images are shown for reference, note the skeletons on each figure.)

activities such as walking and running. Despite extensive research on HPE, the related and equally important task of human pose correction has received limited attention.

There is often confusion between the tasks of *pose refinement*, *pose transition*, and *pose correction*. *Pose refinement* focuses on improving the accuracy of initial pose estimates through iterative adjustments. *Pose transition* involves creating smooth transformations between consecutive poses to ensure continuity in motion sequences. In contrast, pose correction specifically addresses and corrects inaccurate or erroneous joint positions to match a desired target pose, focusing on rectifying discrepancies in joint positions. Fig. 3 highlights the difference between these tasks.

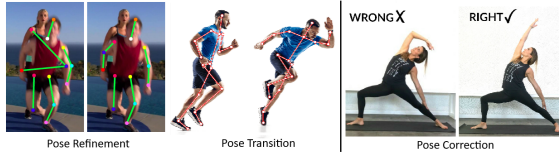


Figure 3. Pose refinement, pose transition vs pose correction task

Although pose classification systems are well-studied, pose correction requires pose detection followed by identification of actionable interventions. Ideally, such a HPC system needs to be explainable to make sure the recommended corrections are actionable. The efforts closest to this work include [10] and [13]; however, [10] is not a user-specific pose correction framework. Our approach is instance-specific and hence aligns more closely with [13], and we compare against them while studying the effectiveness of our proposed framework.

Unlike [13], which focuses solely on generic correction through angles, our proposed framework, PoseGuru Fig. 2, adopts a more comprehensive approach. PoseGuru generates corrected landmarks that form a user-specific reference pose, directly addressing the subjective and anatomical variability inherent in human anatomy. These landmarks are integral to numerous downstream applications, such as fitness apps and online physical therapy platforms, where users can visually compare their initial incorrect pose to the corrected reference pose. This visual comparison empowers users to progressively refine their form and achieve their target pose through self-correction.

Moreover, the landmarks serve as vital reference points for ongoing adjustments and tracking an individual’s progress, offering a tangible and anatomically meaningful representation that enhances the effectiveness of assessing and refining poses in practical settings. To support the evaluation of PoseGuru, we introduce two new derived datasets in this work: YogaHPC and Pilates32+P, which include correct poses, incorrect poses as well as class-specific exemplars corresponding to the Yoga-82 [38] and Pilates32 [12] datasets respectively.

Additionally, our approach employs instance-based explanations [19, 20], where exemplars aid class-specific correction, and counterfactuals iteratively refine incorrect poses while respecting user-specific constraints. This case-based reasoning enhances interpretability.

We focus on 2D image-based pose correction in this work, prioritizing scalability across a wide range of applications where users may only have access to a basic camera, such as on edge devices and mobile platforms. While 3D imaging can offer additional depth information, its scalability is limited by cost and accessibility. Moreover, the datasets used for our evaluation—YogaHPC and Pilates32+P—are 2D-based, further supporting our choice of a 2D approach. In our setup, the camera angle remains consistent, either frontal or side-view (profile), providing reliable conditions for evaluating pose transitions effectively without concerns of variability caused by changing perspectives.

We also introduce two new derived metrics to evaluate HPC task: MPIJAD (Mean per Incorrect Joint Absolute Deviation) and PCIK (Percentage of Corrected Incorrect Joints). Unlike traditional metrics such as MAD and PCK, which evaluate both correct and incorrect joints, these metrics focus solely on the correction of misaligned joints, providing a more targeted and accurate assessment of pose correction effectiveness. We note that this work is conducted entirely without the use of manual skeleton annotations, relying solely on noisy annotations derived from existing pose estimators. Additionally, our framework builds upon these estimators, allowing for the seamless integration of newer off-the-shelf methods, making our approach highly generalizable. At the core of PoseGuru is a novel pose correction optimization algorithm that explores the objective space of landmarks

under the constraints provided by the exemplar and given incorrect pose. We leverage algorithmic recourse (AR) for this purpose. This algorithm is simple and easy to implement, and allows for modification to include application-specific Lagrangian constraints. Below is a summary of our key contributions:

- We propose a novel Human Pose Correction (HPC) approach, generating user-specific reference landmarks for the target pose through algorithmic recourse. This allows users to compare their incorrect pose with the corrected landmarks for real-time feedback and progress tracking.
- Our framework provides user-specific pose corrections grounded in anatomical constraints. By taking into account individual physical limitations, such as restricted mobility of the limbs, personalized and biomechanically feasible adjustments are ensured.
- We propose two new derived datasets, YogaHPC and Pilates32+P, which include correct poses, class-specific exemplars, as well as incorrect poses generated by perturbing the landmarks of correct ground truth poses while respecting anatomical constraints. Given the limited research in pose correction, these datasets can help advance methods in this field.
- We propose two derived metrics, MPIJAD (Mean Per Incorrect Joint Absolute Deviation) and PCIK (Percentage of Corrected Incorrect Joints), specifically designed to evaluate misaligned joints. With limited prior work in this area, these metrics provide a more focused assessment.

2. Related Work

Pose Estimation, Classification and Refinement. *Pose estimation* involves determining the pose of an individual from visual data. A wide range of techniques ranging from traditional methods using hand-crafted features to modern approaches employing deep learning have been proposed for this task over the years [2, 4, 5, 8, 14, 15, 17, 24, 27, 31, 37, 44, 46, 50]. *Pose classification* methods categorize poses into predefined classes as required for different applications. Examples of past work include yoga pose classification [6, 16, 21, 22, 29, 34, 42], facial pose classification [11, 26, 33], gesture recognition [3, 9, 47, 49], human-object interaction [23, 40, 41, 45, 48] and surveillance [1, 7, 35, 39]. The methods largely involve the use of vision-based neural network models to classify the poses. Pose classifiers however are not capable of localizing and correcting pose errors.

Pose refinement methods aim to enhance the precision of an estimated pose. Recent works in pose refinement include [18, 25, 30, 36, 43]. Pose refinement is generally generic and not specific to given pose categories. We note that the term ‘pose correction’ in [30, 36] refers to refining the output of a pose estimation model using MoCap and music respectively, which is more aligned with pose refinement. In contrast, pose correction requires more substantial modifications to rectify significant errors with respect to a target pose.



Figure 4. Both the above images are classified as *reverse warrior* pose by pose classifiers, which overlook errors while predicting the pose category.

Pose Correction. Pose correction, an aspect of human-computer interaction, focuses on identifying and rectifying errors in body part alignment with respect to a target pose. Very limited efforts have been seen for this problem so far. Related works such as [6, 10] provide corrective feedback, but lack detailed mechanisms for explaining how the corrections are determined. These methods rely on having a correct pose of the same individual as a reference, which may be impractical in real-world scenarios. These methods have also not been evaluated on cases where the corrected pose belongs to a different person. While [12] used traditional methods like K-Nearest Neighbor (KNN) and joint angle likelihood to bring in explainability of error localization, it does not explicitly address pose correction. CARE [13] is the closest effort to our objectives towards human pose correction. However, this approach is limited as it relies solely on joint angles. Consequently, it does not provide direct corrected landmarks, does not consider the user’s anatomical constraints, and proposes corrections based on the training data distribution of the target class. In contrast, our work herein introduces a novel approach with landmark-based exemplar-guided algorithmic recourse optimization. Our method uses landmarks to retain the user’s body constraints and incorporates an exemplar-based approach for the correction task. Utilizing landmarks provides a more intuitive understanding of pose adjustments compared to mere joint angles.

3. PoseGuru: Proposed Framework

Simple vision-based pose classifiers, limited to fixed pose categories, often overlook local errors—still predicting the correct class even when parts of the pose are misaligned.

As shown in Fig. 4, pose classification techniques often miss local pose errors, correctly predicting the category despite misalignments—making them inadequate for correction tasks. Existing methods like [13], which rely on pose angles, offer only coarse corrections and fail to address subtle inaccuracies. To overcome this, we propose PoseGuru (Fig. 5), a landmark-based algorithmic recourse optimizer capable of delivering fine-grained corrective feedback.

3.1. Overview of Proposed Framework

Given the incorrect pose landmarks set x' , our objective is to generate a corrected pose landmarks set x^* that adheres to the user’s anatomical integrity and joint fidelity. To this end, we use an exemplar x^{ex} from the desired pose class and a

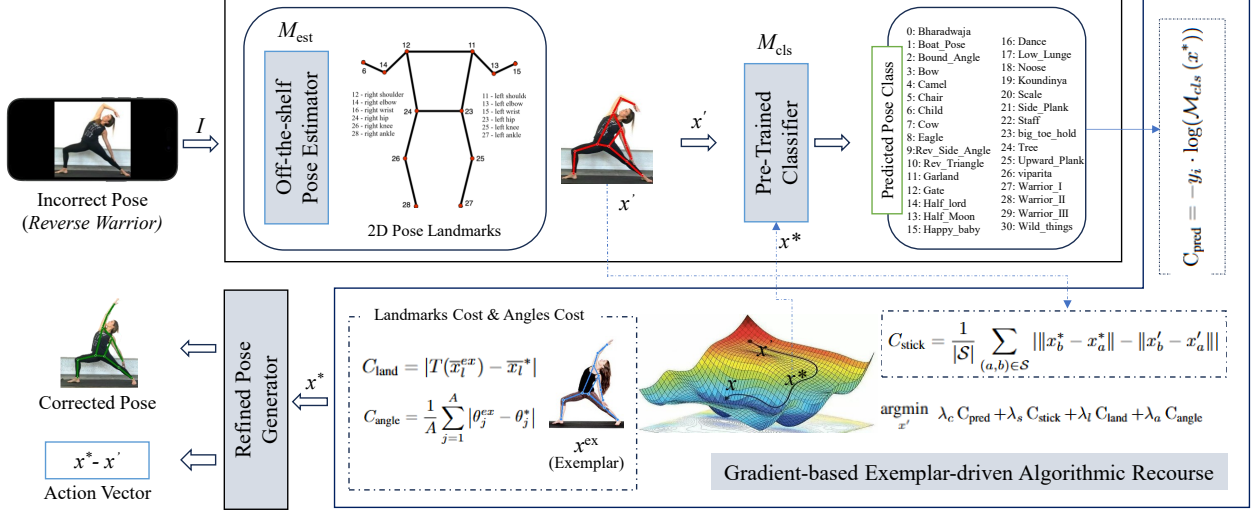


Figure 5. **PoseGuru Framework:** (Top:) Pre-processing phase – Extraction of pose x' with relevant landmarks of an individual performing an incorrect *reverse warrior* pose in front of a monocular camera, followed by the prediction of the x' pose class. (Bottom:) Exemplar-guided algorithmic recourse phase involving an optimizer with four costs: (i) prediction cost C_{pred} , defined as cross-entropy loss of candidate counterfactual x^* , (ii) stick length cost C_{stick} , defined as mean stick length deviation between x^* and x' , (iii) landmark cost C_{land} , defined as mean landmark deviation between x^* and the Procrustes-transformed exemplar $T(x_i^{\text{ex}})$, and (iv) angular cost C_{ang} , defined as mean angular deviation between x^* and x^{ex} . See Sec. 3.4 for more details. Refinement phase – Landmarks with minimal change in x^* are reset to their original positions in x' , generating the corrected pose and sparse action vector.

pre-trained pose classifier $\mathcal{M}_{\text{cls}}$. As illustrated in Fig. 5, our proposed PoseGuru framework operates in three phases:

- **Pre-processing:** PoseGuru processes an RGB image of an individual with an incorrect body pose and estimates its landmarks using an off-the-shelf pose estimator. These are denoised and used to train a human pose classifier (described in Sec. 3.2).
- **Exemplar-guided Algorithmic Recourse:** The estimated landmarks are fed into a novel exemplar-guided algorithmic recourse based optimizer. This optimizer incorporates anatomical constraints, ensuring that all stick lengths in the user's pose are consistently maintained to respect the individual's natural body structure. The algorithmic recourse process utilizes a pre-trained pose classifier and an exemplar pose from the desired pose class to accurately guide the correction.
- **Pose refinement:** The optimized pose is further refined using a "one-body-element-at-a-time" approach wherein we reset the non-peak element deviations to fix the limb or torso that is going wrong. Here a body element could be any of the four limbs or the torso.

Finally, a sparse action vector is generated, providing actionable interventions for the user to correct their pose. We now discuss the different modules required to implement the above phases individually.

3.2. Human Pose Classifier

Given a dataset $D = \{(I_i, y_i)\}_{i=1}^N$, where I_i represents an RGB image of a human pose and $y_i \in C$ is the corresponding pose class label from a set of categories $C =$

$\{y_1, y_2, \dots, y_N\}$, our objective is to train a pose classifier $\mathcal{M}_{\text{cls}}$ that can identify the pose class from the RGB images. To achieve this, we first utilize an off-the-shelf pose estimator $\mathcal{M}_{\text{est}}$ to extract the set of 2D pose landmarks x_i . Since $\mathcal{M}_{\text{est}}$ is not specifically trained on our dataset, the resulting landmarks may be noisy, i.e.:

$$x_i^p = \mathcal{M}_{\text{est}}(I_i) + \epsilon_i$$

where ϵ_i denotes the noise introduced in the landmark extraction process. To ensure robustness, we apply a filtering function f_{filter} to remove landmarks irrelevant to real-world exercise poses (e.g., eyes, mouth, ears). This results in a filtered set of landmarks $x_i \subseteq x_i^p$, where only relevant keypoints for pose classification are retained:

$$x_i = f_{\text{filter}}(x_i^p) \quad (1)$$

From the original dataset D , we construct a new dataset $D_L = \{(x_i, y_i)\}_{i=1}^N$, consisting of filtered pose landmarks corresponding to each image I_i . This dataset D_L is subsequently used to train a neural network classifier $\mathcal{M}_{\text{cls}}$ to predict the pose category: The classifier $\mathcal{M}_{\text{cls}}$ is trained to minimize the classification loss $\mathcal{L}_{\text{cls}}$ in the training set:

$$\mathcal{L}_{\text{cls}} = \frac{1}{N} \sum_{i=1}^N L(\mathcal{M}_{\text{cls}}(x_i), y_i), \quad (2)$$

where L is the loss function of cross-entropy.

3.3. Exemplar Pose Selection

For each class in the pose landmark dataset D_L , we first select the top-5 poses from the training data using the pose

classifier $\mathcal{M}_{cls}$. From these five, we manually select the best exemplar pose by visual inspection, which aids in generating counterfactual explanations for incorrectly formed poses. Unlike existing pose correction methods that focus solely on joint angles, our approach incorporates both the pose class and the body orientation (left/right) of the incorrect pose. This is important because selecting an exemplar with a mismatched orientation (e.g., using a left-oriented exemplar to correct a right-oriented pose) significantly impairs the correction process. Thus, for each class y_i in the dataset D_L , we choose a set of exemplars $\mathcal{E}_{y_i}$ (with $y_i \in C$).

3.4. Counterfactual Pose Correction using Algorithmic Recourse

We model the correction of the erroneous pose as a counterfactual generation problem. Counterfactual explanations identify minimal changes to the input in order to change the output of a function to a target value. Formally, for a normalized input feature x' , the goal is to find a minimal perturbation x^* such that the classifier $\mathcal{M}_{cls}$ outputs a target class y_i .

$$\begin{aligned} \underset{x'}{\operatorname{argmin}} \quad & \|x' - x^*\| + L(\mathcal{M}_{cls}(x^*), y_i) \\ \text{subject to } & x^* \in \mathcal{X} \end{aligned} \quad (3)$$

where $\mathcal{X}$ is a set of domain-specific constraints related to plausibility. Here, L is the cross-entropy loss (we use this over the hinge loss used in [28] since our framework benefits from having output probabilities which can be used for fine-grained pose correction. For real-world pose correction, the generated counterfactuals should not only be plausible, but also be actionable for the user. To this end, we utilize an algorithmic recourse formulation with the below four terms:

- *Prediction Cost*: Measures the cross-entropy loss between predicted probabilities and ground truth label to align the candidate x^* with the desired prediction:

$$C_{\text{pred}} = -y_i \cdot \log(\mathcal{M}_{cls}(x^*)) \quad (4)$$

- *Stick Length Cost*: Evaluates the L1 distance (normalized) between corrected pose stick vectors and incorrect pose stick vectors to adjust pose stick lengths:

$$C_{\text{stick}} = \frac{1}{|\mathcal{S}|} \sum_{(a,b) \in \mathcal{S}} \left| \|x_b^* - x_a^*\| - \|x_b' - x_a'\| \right| \quad (5)$$

where $\mathcal{S}$ is the set of all possible joint pairs, x_a^*, x_b^* are joint vectors in the corrected joints representation, and x_a', x_b' are joint vectors in the incorrect joints representation.

- *Landmarks Cost*: Measures the L1 distance between centered counterfactual x^* , denoted by $\overline{x^*}$, and the centered and transformed exemplar pose $T(x_i^{ex})$ using the Procrustes distance with Singular Value Decomposition (SVD), ensuring structural alignment.

$$C_{\text{land}} = |T(\overline{x_l^{ex}}) - \overline{x_l^*}| \quad (6)$$

- *Angles Cost*: Calculates the L1 distance between angles of key joints in the exemplar pose and the candidate pose, ensuring human anatomical plausibility [32]. A is the number of angles.

$$C_{\text{angle}} = \frac{1}{A} \sum_{j=1}^A |\theta_j^{ex} - \theta_j^*| \quad (7)$$

We combine the above terms to formulate the algorithmic recourse optimization problem as:

$$\underset{x^*}{\operatorname{argmin}} \lambda_c C_{\text{pred}} + \lambda_s C_{\text{stick}} + \lambda_l C_{\text{land}} + \lambda_a C_{\text{angle}} \quad (8)$$

Notice that the feasibility constraints are absorbed in the Lagrangian terms for exemplar-guided angles and landmarks cost while stick length cost takes care of actionability. Here, $\lambda_c, \lambda_s, \lambda_l$ and λ_a are coefficients chosen via hyperparameter tuning. In practice, we solve this optimization using Stochastic Gradient Descent with Adam optimizer and Nesterov momentum to obtain the optimized pose vector. Any user-specific limitations in range of motion can also be added as feasibility constraints, if available.

3.5. Pose Refinement

We introduce a post-processing pose refinement step that focuses on local corrections, instead of full-body corrections. To implement this, the optimized pose x^* is compared to the initial incorrect pose x' by calculating the L1 norm deviation for each body segment: left arm (x_{la}), right arm (x_{ra}), left leg (x_{ll}), right leg (x_{rl}), and torso (x_t). The L1 norm deviation for each segment (seg) is defined as:

$$\delta_{\text{seg}} = \|x_{\text{seg}}^* - x_{\text{seg}}'\|_1, \quad \text{for seg} \in \{la, ra, ll, rl, t\} \quad (9)$$

The segment with the maximum deviation is identified as:

$$\text{max_deviation} = \arg \max_{\text{seg}} (\delta_{la}, \delta_{ra}, \delta_{ll}, \delta_{rl}, \delta_t) \quad (10)$$

For the segment with the maximum deviation, its corresponding value in x^* is retained, while the remaining segments are reset to their original values from x' . The corrected pose is computed as:

$$x_{\text{final, seg}}^* = \begin{cases} x_{\text{seg}}^* & \text{if } \delta_{\text{seg}} = \text{max_deviation}, \\ x_{\text{seg}}' & \text{otherwise.} \end{cases} \quad (11)$$

This ensures that only the body segment with the greatest deviation—whether it's the left arm, right arm, left leg, right leg, or torso—is adjusted, while others are restored to their original positions, allowing for targeted correction of the misaligned parts. Our approach creates incorrect data (see Sec. 3.6) by perturbing a single body segment. However, PoseGuru can handle multi-segment perturbations by iteratively passing the refined pose back through the framework for each segment needing correction.

Our overall pose correction algorithm is summarized in Algorithm Algorithm 1 (and Fig. 5).

Algorithm 1 Pose Correction using PoseGuru

Require: Incorrect pose image I , pose classifier $\mathcal{M}_{cls}$, pose estimator $\mathcal{M}_{est}$, target class y , tolerance threshold $\mathcal{T}$

- 1: Initialize weight parameters $\lambda_c, \lambda_s, \lambda_l, \lambda_a$
- 2: $x^p \leftarrow \mathcal{M}_{est}(I)$ // Obtain pose landmarks
- 3: $x' \leftarrow \text{filter_landmarks}(x^p, 12)$ // Filter relevant landmarks
- 4: $y' \leftarrow \mathcal{M}_{cls}(x')$ // Obtain pose category
- 5: Compute $C_{pred}, C_{angle}, C_{stick}, C_{land}$
- 6: $C(x) \leftarrow \lambda_c \cdot C_{pred} + \lambda_s \cdot C_{stick} + \lambda_l \cdot C_{land} + \lambda_a \cdot C_{angle}$
- 7: $x^* \leftarrow \arg \min_{x'} C(x)$ // Generate counterfactual explanation
- 8: Split x^* into body elements: $x^* = \{x_{la}^*, x_{ra}^*, x_{ll}^*, x_{rl}^*, x_t^*\}$ // left arm, right arm, left leg, right leg, torso
- 9: $\delta_i \leftarrow \|x_i^* - x'_i\|$, for each $i \in \{la, ra, ll, rl, t\}$ // Compute deviation for body element
- 10: $i_{max} \leftarrow \arg \max \{\delta_{la}, \delta_{ra}, \delta_{ll}, \delta_{rl}, \delta_t\}$ // Identify element with maximum deviation
- 11: **for** each $i \in \{la, ra, ll, rl, t\}$ **do**
- 12: **if** $i \neq i_{max}$ **then**
- 13: $x'_i \leftarrow x'_i$ // Reset all except max deviation element
- 14: **end if**
- 15: **end for**
- 16: $x'_{i_{max}} \leftarrow x^*_{i_{max}}$ // update the incorrect pose for the correct limb
- 17: $v_a \leftarrow x^* - x'$ // Compute action vector
- 18: **return** x^*, v_a Return Counterfactual explanation and action vector

3.6. Generating Incorrect Poses

To effectively evaluate a pose correction system, we require a set of incorrect poses. An incorrect pose $x' \in \tilde{D}_L$ deviates significantly from its corresponding value in the ground truth pose x for a given class $y_i \in \mathcal{C}$. Although incorrect pose landmark vectors can be randomly generated, many configurations are biomechanically unattainable by humans due to joint angle and movement constraints.

We hence generate an incorrect pose dataset $\tilde{D}_L$ by perturbing the M landmarks $l_1, l_2, \dots, l_M$ of the ground truth x within biomechanically plausible constraints. Specifically, each selected joint is perturbed by an angle θ_{pert} , constrained by maximum allowable deviations θ_{max} , where:

$$\theta' = \theta^{gt} + d_i \cdot \theta_{pert} \quad (12)$$

Here, $d_i \in \{-1, 1\}$ determines the perturbation direction, where $d_i = -1$ for clockwise (negative angle) and $d_i = 1$ for counterclockwise (positive angle).



Figure 6. **Incorrect Poses:** Generation of incorrect poses. Correct pose is modified under bio-mechanically plausible constraints (shown in red)

We impose the following specific constraints during perturbation [32]: (i) **Avoid Unattainable Angles:** Joint angles must respect limits, defined by: $\theta_{shoulder} \in [0^\circ, 180^\circ]$, $\theta_{hip} \in [0^\circ, 120^\circ]$ and $\theta_{elbow}, \theta_{knee} \in [30^\circ, 180^\circ]$; (ii) **Limit**

Perturbations of End Joints: Perturbing distal joints (ankles, wrists) requires adjusting proximal joints (knees, elbows); we hence limit perturbations to 8 joints instead of 12; and (iii) **Maintain Dependent Joint Relationships:** Adjust shoulder or hip angles with corresponding changes in the elbow/knee and wrist/ankle to maintain natural movement.

We illustrate a few samples of the incorrect pose in Fig. 6.

4. Experiments and Results

We performed detailed experiments (Table 1) and ablation studies (Table 2) to study the effectiveness of the proposed framework.

Datasets. We used two datasets, YogaHPC and Pilates32+P, to evaluate the effectiveness of PoseGuru. These datasets are derived from the Yoga-82 [38] and Pilates32 [12] datasets to study the proposed setting. Our derived datasets will be made publicly available for further research and reproducibility. *YogaHPC:* A subset of the Yoga-82 dataset with 31 diverse classes that allow for pose correction are considered in this dataset. The classes include five with severe self-occlusion, fifteen with moderate self-occlusion and eleven occlusion-free poses. Our YogaHPC dataset consists of 3658 training and validation samples split in 80:20 and 2613 test samples. Additionally, the incorrect pose dataset includes 150 training samples and 150 test samples, created by perturbing ground truth poses as detailed in Sec. 3.6. *Pilates32+P:* This is built on the Pilates32 [12] dataset to include pose information and incorrect poses for correction. The dataset, which contains 32 Pilates exercises, includes 1890 training and validation samples split in 80:20 and 266 test samples, with an additional incorrect pose dataset comprising 64 training samples and 64 test samples, generated by perturbing ground truth poses as in Sec. 3.6.

Each dataset in turn includes an image dataset, a pose dataset of 12 normalized landmarks, and an incorrect pose dataset of 12 normalized landmarks. Each normalized Landmark represents a point in 2D space with x, y coordinates. x and y are normalized to $[0, 1]$ by image width and height.

Evaluation Metrics. Considered there has been very little previous work on pose correction, we introduce two refined metrics – MPIJAD and PCIK – to effectively evaluate PoseGuru’s performance.

MPIJAD: We consider PCP (Percentage of Correct Poses) at Mean Absolute Deviation (MAD) from [13], which measures the proportion of poses where the MAD between ground truth and corrected landmarks is below a threshold $\mathcal{T}$. However, MAD includes deviations from both correct and incorrect landmarks, which can dilute the impact of larger errors in misaligned joints. To address this, we use *MPIJAD (Mean Per Incorrect Joint Absolute Deviation)*, a metric that focuses on incorrect joints, providing a more accurate evaluation of how well PoseGuru corrects inaccuracies on

misaligned joints. We use a threshold of $\mathcal{T} = 0.1$ across the experiments in this work. (Results with variations of this threshold are presented in the supplementary Tab. 5.)

PCIK: In addition, the widely used PCK (Percentage of Correct Keypoints) measures the proportion of keypoints within a threshold distance from their ground truth positions and is commonly employed in pose estimation. However, since PCK includes keypoints that were already correct, it can obscure the effectiveness of the correction process. To address this, we propose *PCIK (Percentage of Corrected Incorrect Keypoints)*, which evaluates only the originally incorrect keypoints, offering a clearer assessment of pose correction effectiveness.

Together, PCP at MPIJAD measures overall pose correction accuracy, while PCIK focuses on individual keypoint corrections in incorrectly aligned keypoints. These metrics provide a more complete evaluation of PoseGuru’s performance, ensuring overall accuracy and targeted improvement.

Experimental Setup/Implementation Details. We employ the BlazePose [2] pre-trained pose estimation model to obtain the pose estimates from both datasets. Additionally, for training pose classifiers, we utilize individual shallow, fully connected neural networks dedicated to each dataset. The architectures for YogaHPC and Pilates32+P are [24, 128, 64, 31] and [24, 256, 128, 64, 32] respectively. These models are trained using a learning rate of 0.01, 1000 steps, and the NAdam optimizer. λ_c , λ_s , λ_l and λ_a where chosen to be 0.1, 0.2, 0.3 and 0.4 respectively. An analysis of these choices are included in the supplementary Sec. A.

Baseline Methods. Considering the paucity of previous work for pose correction, we compare our method’s performance against one known method, CARE [13] and 5 other methods that we believe can be applied to this problem : (i) ANN Regressor, a neural network-based regressor with 4 layer architecture that predicts the corrected pose from an incorrect pose vector; (ii) Exemplar, which directly uses the exemplar for pose correction; (iii) Procrustes Exemplar, which incorporates Procrustes distance to align the centered exemplar with the centered

Results. Tab. 1 shows our quantitative results. Evidently, PoseGuru outperforms all baselines, especially on the PCP MPIJAD and PCIK metrics, which demonstrate its strong targeted correction performance. Since the number of incorrect landmarks can be small in localized corrections, MAD performs well, but PCP at MAD may not be as meaningful patterns as PCP at MPIJAD; the latter ignores already correct joints and focuses on MAD per incorrect joint. In particular, we note that PoseGuru outperforms CARE [13] on both datasets. Similarly, the standard PCK metric is inadequate for pose correction because it includes correct joints. PoseGuru outperformed CARE on PCIK. Exemplar and Procrustes exemplar baselines also do well here, showcasing the need for an exemplar-based approach. CARE’s under-

	Method	PCP MAD $\uparrow$	PCP MPIJAD $\uparrow$	PCK $\uparrow$	PCIK $\uparrow$
YogaHPC	Centroid	0.98	0.40	0.76	0.26
	Medoid	0.97	0.46	0.78	0.31
	Exemplar	<u>0.99</u>	<u>0.65</u>	0.82	0.41
	Proc. Exemplar	0.93	0.71	0.85	0.74
	ANN Regressor	<u>0.99</u>	0.21	0.68	0.45
	CARE [13]	1.00	0.20	<u>0.86</u>	<u>0.76</u>
	PoseGuru	<u>0.99</u>	0.71	0.89	0.82
Pilates32+P	Centroid	1.00	0.34	0.72	0.15
	Medoid	1.00	0.34	0.73	0.15
	Exemplar	1.00	0.44	0.80	0.25
	Proc. Exemplar	0.98	<u>0.65</u>	0.88	<u>0.43</u>
	ANN Regressor	1.00	0.26	0.70	0.14
	CARE [13]	1.00	0.32	<u>0.89</u>	0.19
	PoseGuru	1.00	0.77	0.90	0.81

Table 1. **Main Results:** PoseGuru consistently outperformed all baselines for YogaHPC and Pilates32+P datasets, especially on the PCP MPIJAD and PCIK metrics, which demonstrate its strong targeted correction performance. **Best results** are in bold and second best results are in underline.

performance may be attributed to the lack of exemplars and stick length consistency. PoseGuru’s robust performance on both YogaHPC and Pilates32+P datasets underscores its generalization ability across diverse pose types. Leveraging exemplars and landmark data helps provide improved anatomical integrity and joint fidelity, while algorithmic recourse optimization efficiently corrects landmarks within user-specific constraints.

Qualitative results are presented in Fig. 2, Fig. 7 and Fig. 8. Fig. 2 shows results on diverse poses from YogaHPC and Pilates32+P datasets, including the exemplar, ground truth pose (in black), incorrect pose (in red), PoseGuru generated counterfactual explanation (CFE) (in green) and finally the corrected pose (also in green). Fig. 7 shows more qualitative results with ground truth, incorrect and corrected poses. Fig. 8 shows the effectiveness of PoseGuru on samples from YogaHPC with significant self-occlusions. PoseGuru performs well even on these poses. Although quantitative explainability measures are valuable, PoseGuru’s qualitative explainability is sufficient, as counterfactuals inherently clarify how poses must change to reach the target class.

5. Ablation Analysis and User Studies

User Study. We conducted a comprehensive user study to evaluate the effectiveness of PoseGuru in correcting poses from the YogaHPC and Pilates32+P datasets. We engaged 4 yoga and pilates experts to rate 8 pose corrections from each dataset, assessing PoseGuru’s performance using a standard 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). This study was aimed at understanding the usefulness of the corrections provided by PoseGuru.

Study Setup: 4 yoga and 4 pilates experts with an average of 9+ years of experience were engaged to evaluate 8 poses each on YogaHPC and Pilates32+P datasets for correction

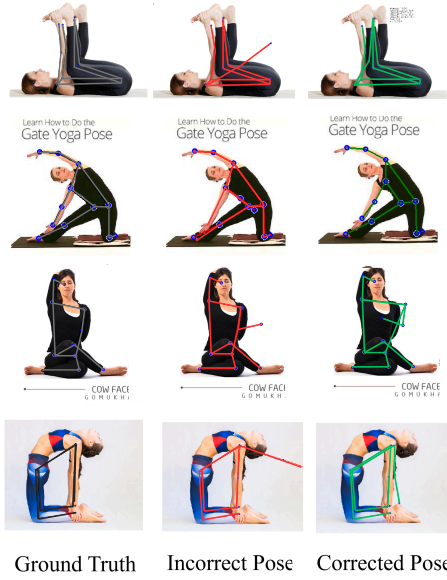


Figure 7. **Sample Qualitative Results.** Ground truth poses (in black), Incorrect poses (in red), PoseGuru-generated corrected poses (in green).

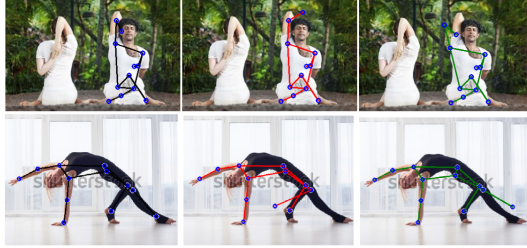


Figure 8. **More Qualitative Results.** PoseGuru on poses with significant self-occlusion. Ground truth poses (in black), Incorrect poses (in red), PoseGuru-generated corrected poses (in green).

analysis. Each expert rated PoseGuru’s correction based on visual assessments of misaligned joints, adherence to biomechanical integrity, and overall alignment with desired target pose. Ratings were given on a 5-point Likert scale.

Results: The average ratings for YogaHPC corrections ranged between 3.00 (Neutral) and 4.25 (Agree), with an overall average rating of 3.65. Considering that these were experts in the field, these scores are high (as also qualitatively stated by these experts in our user sessions). While the system performed well in poses with clear misalignments, minor errors in joint positioning were occasionally missed, leading to a few lower ratings. The ratings for Pilates32+P showed higher ratings overall, with an average correction rating of 3.81. Experts were particularly impressed by PoseGuru’s accuracy in aligning core and extremity positions. Lower ratings were given for poses where small misalignments in joints were present. Experts noted that while the system performs well, there is room for improvement in handling more subtle joint adjustments in future iterations. This provides scope for future extensions of this work.

Effect of Terms in Objective Function. Tab. 2 shows an ablation study of the cost terms in our algorithmic recourse

Metric	Objective	Thresholds ($\mathcal{T}$)				
		0.02	0.04	0.06	0.08	0.10
PCP (MPIJAD $\leq \mathcal{T}$)	Pred	0.01	0.01	0.05	0.12	0.19
	Pred + Stick	0.00	0.01	0.03	0.14	0.22
	Pred + Stick + Land	<u>0.13</u>	<u>0.27</u>	<u>0.41</u>	0.49	0.58
	Pred + Stick + Ang	0.11	<u>0.27</u>	0.39	<u>0.51</u>	<u>0.59</u>
	PoseGuru	0.15	0.39	0.56	0.66	0.71
PCIK (within $\mathcal{T}$)	Pred	0.42	0.51	0.61	0.66	0.69
	Pred + Stick	0.58	0.65	<u>0.71</u>	0.73	0.75
	Pred + Stick + Land	0.52	0.62	0.69	0.75	<u>0.80</u>
	Pred + Stick + Ang	<u>0.54</u>	0.61	0.69	<u>0.76</u>	<u>0.80</u>
	PoseGuru	<u>0.54</u>	<u>0.63</u>	0.72	0.78	0.82

Table 2. **Ablation Study on Objective Terms.** MPIJAD & PCIK results for various combinations of cost terms. Clearly, our overall framework objective performs the best. **Best results** in bold, 2nd-best results underlined.

optimization. We observe that the objective that uses all terms consistently performs better across all considered metrics and thresholds, corroborating our overall method design.

Effect of Choice of Threshold Values. We conduct a detailed study with different threshold values for the MPIJAD metric on the Pilates32+P dataset. The results are shown in the supplementary Tab. 5 due to space constraints. They demonstrate that PoseGuru outperforms baselines across threshold choices.

Effect of Procrustes Transformation of Exemplar: We also studied PoseGuru and its variant with and without Procrustes transformation of the exemplar. The results, shown in the supplementary Tab. 4, demonstrate that the Procrustes transformation improves the performance of our method across all thresholds by 10-25% for PCP MPIJAD and PCIK.

More qualitative results are included in the supplementary Fig. 9 and Fig 10.

6. Conclusions and Future Work

We introduced a novel approach to Human Pose Correction (HPC) by generating user-specific reference landmarks for the target pose through adaptable AR, allowing real-time feedback and progress tracking. Our framework, PoseGuru, corrects misaligned poses using an exemplar, preserving anatomical integrity and joint fidelity without requiring a pre-existing correct pose of the same individual. We also developed two new derived datasets, YogaHPC and Pilates32+P, which fill a crucial gap in this under-researched field. Additionally, we introduced two derived metrics, MPIJAD and PCIK, which focus on misaligned joints, providing a more precise evaluation than traditional metrics. PoseGuru consistently outperformed existing methods across both datasets, demonstrating generalization and effective correction across diverse pose types. Future directions of our work could include 3D-based pose correction, further integration and evaluation of user-specific mobility constraints, and improving correction techniques for video sequences.

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